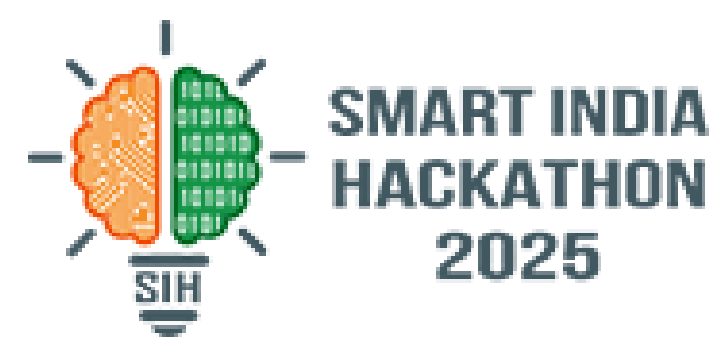


SMART INDIA HACKATHON 2025



Problem Statement ID – SIH25197

Problem Statement Title - AI and ML enabled video analysis and interpretation

Theme - Miscellaneous

PS Category - Software

Team ID - 115740

Team Name - HeisenStack



PROPOSED SOLUTION

- Unified AI-Driven Video Intelligence Platform
- Automated Object, Face, and Trait Recognition
- Intelligent Tracking and Re-Identification
- Adaptive Analysis and Alert Mechanism
- Scalable, Modular, and Secure Architecture



REAL WORLD IMPACT

- Enhanced Situational Awareness
- Faster Threat Detection and Response
- Optimized Use of Existing Infrastructure

INNOVATION

- Hybrid Edge–Cloud AI Architecture
- Unified Multi-Source Intelligence System
- Adaptive AI with Programmable Rules

TECH STACK

Video Ingestion

- OpenCV
- FFmpeg
- asyncio / threads

ByteTrack for multi-object tracking

torchreid (OSNet) for Re-ID

FastAPI for backend

React.js for frontend

PostgreSQL for database

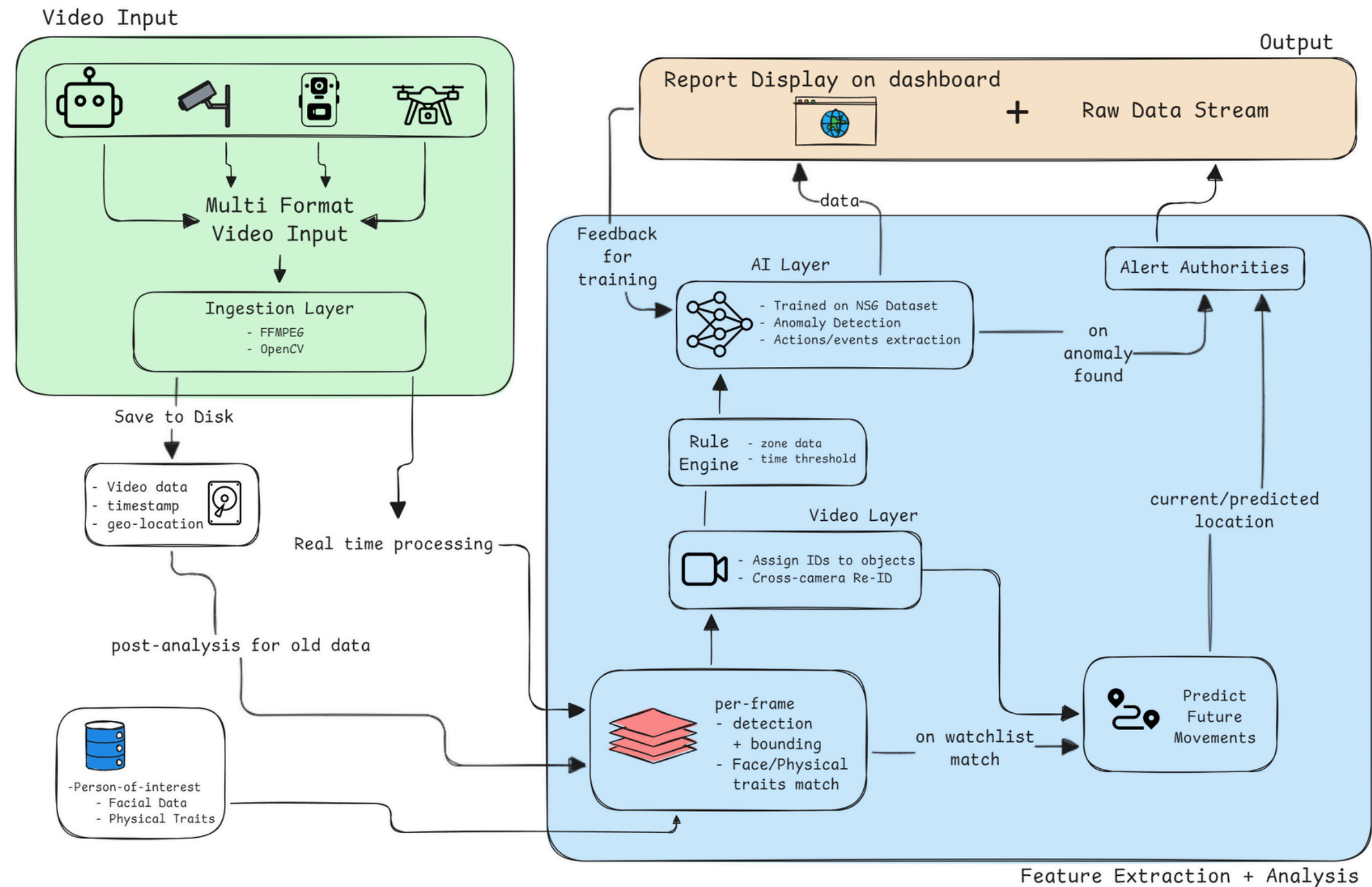
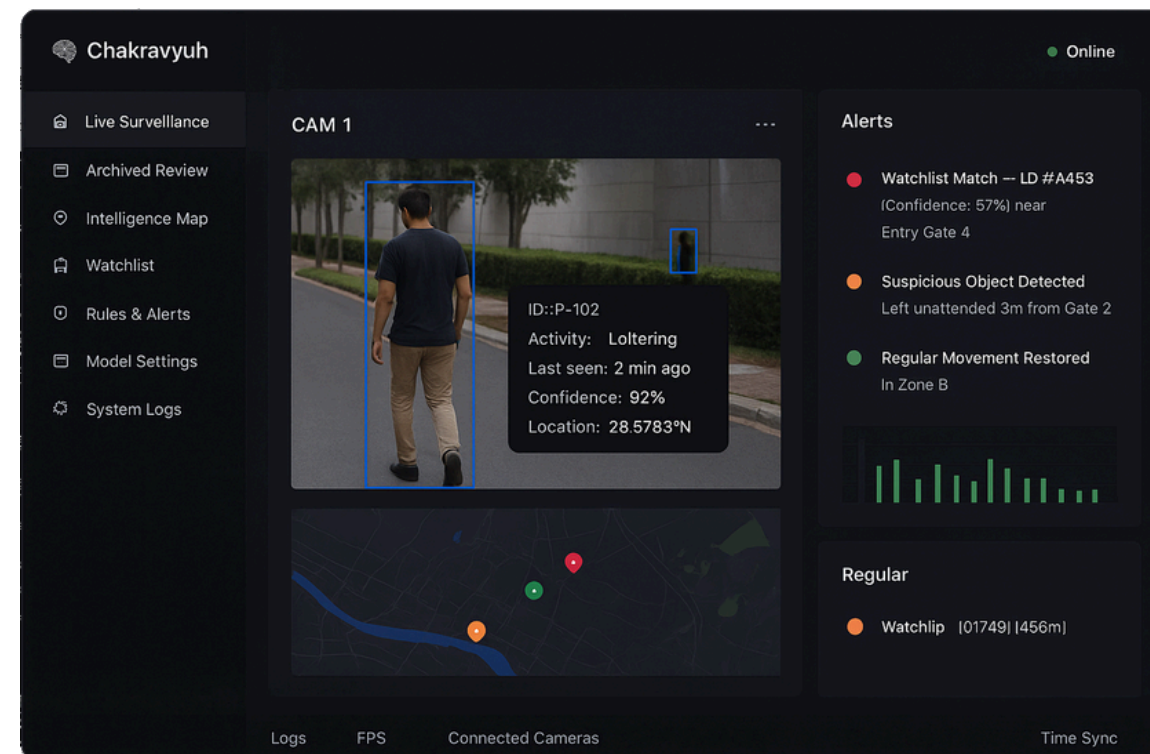
Feature Extraction & AI Layer

- YOLOv8
- PyTorch

Facial and Physical Trait Detection

- InsightFace
- OpenPose
- FAISS

MOCKUPS



Feasibility Analysis



TECHNICAL

Proven CV/ML stack, scalable, modular



FINANCIAL

Open-source tools, no hardware upgrades, GPU cost only



MARKET

High demand (defense, smart cities), cost-effective



OPERATIONAL

User-friendly dashboards, integrates with legacy systems

Potential Challenges & Risks



TECHNOLOGY

Latency, accuracy issues, compute-heavy



COST

Server/GPU cost, retraining overhead



ADOPTION

Competition, adoption trust issues



EXECUTION

Training needs, privacy compliance

Strategies to Overcome



METHODS / PRINCIPLES

Edge+Cloud, Agile rollout, Async processing, Privacy-by-Design



STRATEGIES

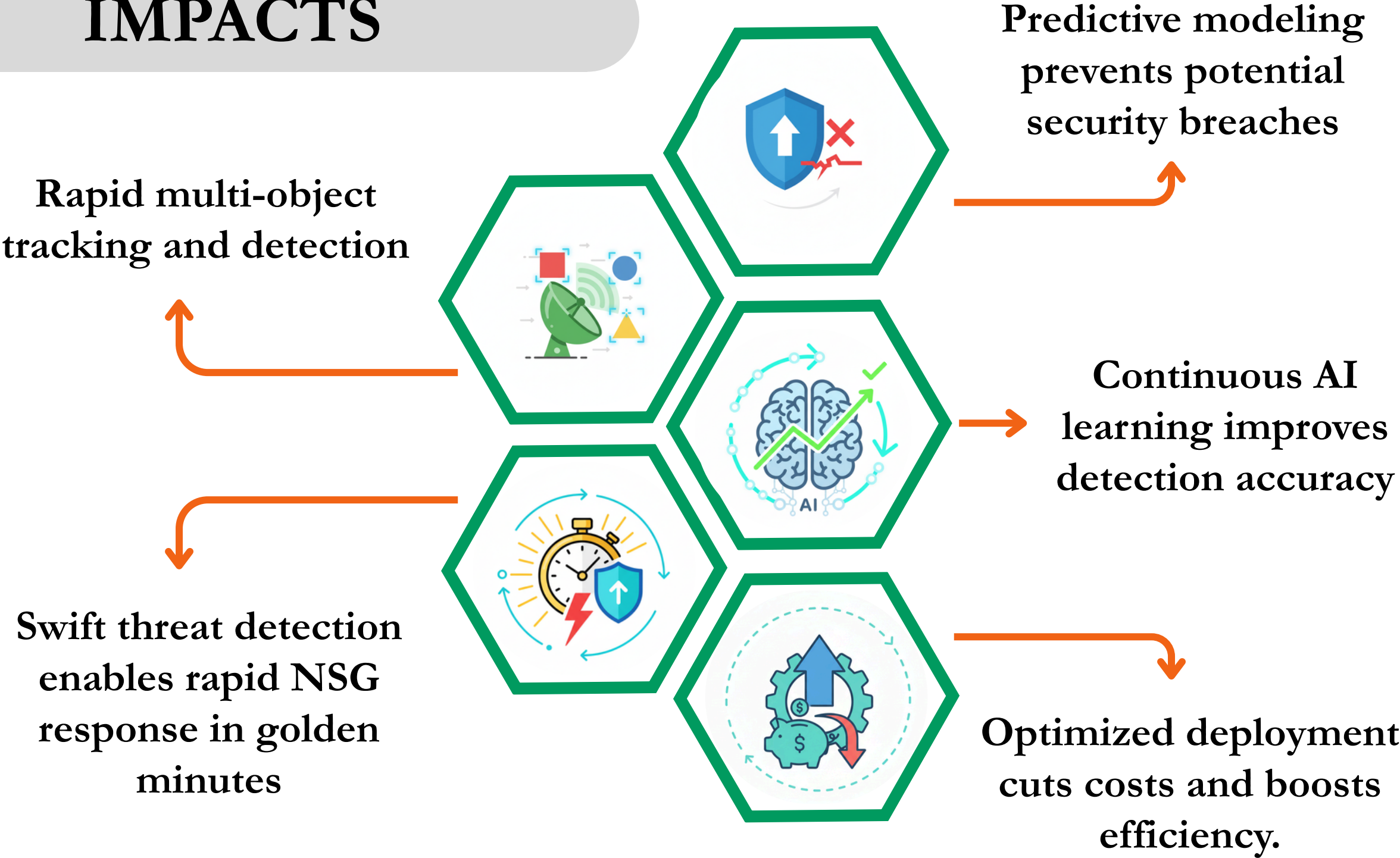
GPU optimization, open-source, pilot deployments, operator training



ALGORITHMS / TOOLS

YOLOv8, ArcFace, FAISS, Custom Rule Engine

IMPACTS



BENEFITS

Public Trust: Builds safety and confidence through ethical surveillance.
Global Standing: Enhances India's image as technologically advanced in counter-terrorism
Economic: Reduces manpower load & reduces both operational and recovery costs.
Strategic: Scalable, modular system adaptable to future threats

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